Dr. Saif Ali Khan

Assistant Professor (Statistics), Directorate of Online Education
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PROFILE

Statistician and researcher with a Ph.D. in Statistics, working at the interface of statistical theory, machine learning and applied data analysis. My research develops and applies statistical and computational methods to model, analyse and predict patterns in real world data across health, agriculture and the social sciences. I have co-authored 10 peer reviewed journal articles, with 51 citations on Google Scholar.

RESEARCH INTERESTS

- Infectious disease modelling and epidemiological forecasting
- Time series analysis and predictive modelling
- Machine learning in public health, agriculture and social sciences

ACADEMIC APPOINTMENTS

Assistant Professor Nov 2025 – Present
Directorate of Online Education, Manipal Academy of Higher Education, Manipal, Karnataka, India

Research Assistant (during Ph.D.) May 2023 – Oct 2024
Department of Statistics, Babasaheb Bhimrao Ambedkar University (BBAU), Lucknow, India

EDUCATION

Ph.D., Statistics 2022 – 2025
Department of Statistics, Babasaheb Bhimrao Ambedkar University, Lucknow, Uttar Pradesh, India
Research focus: mathematical, statistical and machine learning approaches to infectious disease dynamics.

M.Sc., Statistics 2018 – 2020
Department of Statistics & Operations Research, Aligarh Muslim University, Aligarh, Uttar Pradesh, India
Research topic: image classification of skin lesions.

B.Sc. (Hons.), Statistics 2015 – 2018
Department of Statistics & Operations Research, Aligarh Muslim University, Aligarh, Uttar Pradesh, India
Research topic: effectiveness of the Choice-Based Credit System.

RESEARCH FUNDING & PROJECTS

- Uttar Pradesh Council of Science and Technology (UPCST)-funded research project on “Statistical modeling for COVID-19 spread using different approaches”

PEER-REVIEWED JOURNAL PUBLICATIONS

Listed in reverse chronological order; author name in bold.

1. Verma, M. K., & **Khan, S. A.** (2026). Estimation of population mean using multivariate stratified sampling: An analysis of child mortality and parental education. *Health Services and Outcomes Research Methodology*. https://doi.org/10.1007/s10742-026-00393-w
2. Verma, M. K., & **Khan, S. A.** (2026). A robust population mean estimation method in stratified sampling using auxiliary attributes under real and simulated populations. *Advanced Modeling and Simulation in Engineering Sciences, 13*(1), 20. https://doi.org/10.1186/s40323-026-00339-4
3. Khan, M. M. A., Batool, T., Ali, A., **Khan, S. A.**, Naeem, M., & Mukarram, M. (2026). Microwave-induced eustress in *Datura innoxia*: Integrated photosynthetic, oxidative, and cellular responses to a low-dose hormetic elicitor. *Biocatalysis and Agricultural Biotechnology*, 104207. https://doi.org/10.1016/j.bcab.2026.104207
4. **Khan, S. A.**, Singh, V., Yadav, S. K., & Akhter, Y. (2026). A comparative state-wise assessment of dengue incidence and deaths in India using annual surveillance data. *Discover Public Health, 23*(1), 981. https://doi.org/10.1186/s12982-026-02396-x

5. **Khan, S. A.**, Yadav, S. K., & Akhter, Y. (2026). Analyzing global threat of lumpy skin disease virus infection: A comparative study of a time series model and machine learning models across 10 high-risk countries. *Current Microbiology, 83*(7), 389. https://doi.org/10.1007/s00284-026-04982-3
6. Singh, V., Hajare, G. C., **Khan, S. A.**, & Kumari, A. (2026). Decoding socio-economic disparities in Uttar Pradesh: A spatio-temporal analysis using Wroclaw Taxonomy and K-means unsupervised machine learning clustering. *Mathematical Population Studies*(1–23). https://doi.org/10.1080/08898480.2025.2612641
7. **Khan, S. A.**, Shukla, A. K., Yadav, S. K., & Vishwakarma, G. K. (2026). Machine learning models for analysis and prediction of optimal egg production. *Quality & Quantity, 60*(1), 1705–1725. https://doi.org/10.1007/s11135-025-02309-1
8. **Khan, S. A.**, Sharma, D. K., Shukla, A. K., Verma, M. R., Sharma, A., & Yadav, S. K. (2025). Traditional and machine learning models for forecasting apple production and cultivation area. *Applied Fruit Science, 67*(5), 402. https://doi.org/10.1007/s10341-025-01639-6
9. Yadav, S. K., **Khan, S. A.**, Tiwari, M., Kumar, A., Kumar, V., & Akhter, Y. (2024). Taking cues from machine learning, compartmental and time series models for SARS-CoV-2 Omicron infection in Indian provinces. *Spatial and Spatio-temporal Epidemiology, 48*, 100634. https://doi.org/10.1016/j.sste.2024.100634
10. Singh, V., **Khan, S. A.**, Yadav, S. K., & Akhter, Y. (2024). Modeling global monkeypox infection spread data: A comparative study of time series regression and machine learning models. *Current Microbiology, 81*(1), 15. https://doi.org/10.1007/s00284-023-03531-6

PROFESSIONAL QUALIFICATIONS

- GATE – Data Science & Artificial Intelligence (2024)
- O Level – National Institute of Electronics & Information Technology (NIELIT) (2022)

TECHNICAL SKILLS

Programming & Software: Python, SPSS
Methods: Epidemiological and compartmental modelling, time series analysis, machine learning, predictive analytics, survey sampling, spatio-temporal analysis, data visualisation

NATIONAL/INTERNATION CONFERENCES/WORKSHOP

1. Research paper presented: **"Predicting Lumpy Skin Disease Outbreak: A Comparative Study of Machine Learning Across Affected Countries"** at **International Conference on Advances in Geometry algebra, Analysis and Artificial Intelligence (ICAGAAAI-2024)**", organized by Department of Mathematics & Statistics, Integral University, Lucknow, India during November 7-9, 2024.
2. Research paper presented: **"A comparative study of Trends for Dengue outbreaks and mortality in India using different Machine Learning models"** at **International Conference on Innovations in Science, Management and Technology (ICISMT)",** Organized By DEPARTMENT OF BUSINESS ADMINISTRATION Chaudhary Devi Lal University Sirsa, Haryana, India in Association with American Institute of Management and Technology (AIMT), USA (An Academic Subsidiary of The Global Knowledge foundation (GKF), USA, Inc.) during August 17-18, 2023.
3. Participated in the "**Workshop on Computational Oncology**", organized by Department of Mathematics, Indian Institute of Technology Patna held during 19th April to 21st April, 2024.

RESEARCH PROFILES

ORCID: 0000-0003-1722-728X | **Google Scholar:** scholar.google.com | **ResearchGate:** researchgate.net
Web of Science: webofscience.com | **Scopus**: scopus.com